



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech VI Semester
End Semester Examination - April / May -2025
MACHINE LEARNING [IT-4402]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the key phases of the Machine Learning life cycle in sequential order (5)
 - A)
 - B) Given a dataset containing missing values, how would you handle them using different imputation techniques? Provide a brief explanation (5)
- 2) Given a dataset with two classes (pass/fail), how would you apply logistic regression to predict the outcome? List the key steps involved. (5)
 - A)
 - B) How do decision trees make predictions? Explain the role of entropy and information gain in constructing a decision tree. (5)
- 3) Compare Chernoff and Hoeffding bounds. How do these bounds help in understanding the concentration of probability distributions? (5)
 - A)
 - B) In an online learning scenario, if an adversary selects the hardest possible examples at each step, what strategies can be used to minimize the worst-case error? (5)
- 4) Explain the role of the "number of clusters (K)" in K-Means? How can you determine the optimal value of K? (5)
 - A)
 - B) Given a high-dimensional dataset, describe the steps involved in applying PCA for dimensionality reduction. (5)
- 5) Suppose you have an MDP where an agent needs to decide the best action at each state. How would you use the Bellman equation to compute the optimal policy? (5)
 - A)
 - B) Explain how LQR optimizes a system's control inputs using a quadratic cost function. (5)

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